

National University of Technology

Machine Learning - CS3503

LAB Submitted to:

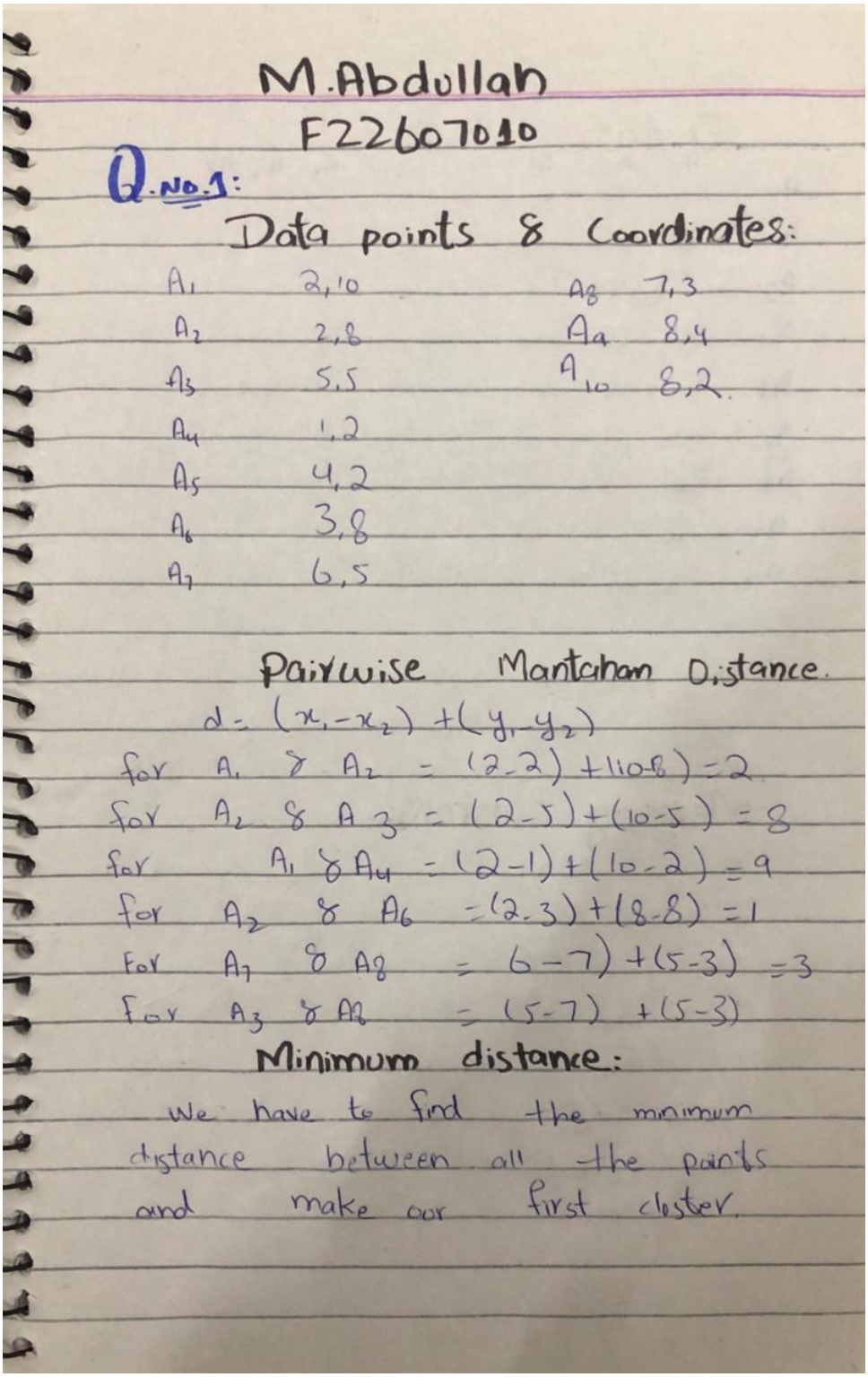
Lec.Hajra Ahmad

Submitted By:

Muhammad Abdullah

Reg. No: **F22607010**
Batch: **Fall AI 2022**
Assignment : **03**

Q.NO.1:



M.Abdullah
F22607010

Q.NO.1:

Data points & Coordinates:

A_1	2,10	A_8	7,3
A_2	2,8	A_9	8,4
A_3	5,5	A_{10}	8,2.
A_4	1,2		
A_5	4,2		
A_6	3,8		
A_7	6,5		

Pairwise Mantahan Distance.

$d = (x_1 - x_2) + (y_1 - y_2)$

for A_1 & A_2 = (2-2) + (10-8) = 2

for A_2 & A_3 = (2-5) + (10-5) = 8

for A_1 & A_4 = (2-1) + (10-2) = 9

for A_2 & A_6 = (2-3) + (8-8) = 1

For A_7 & A_8 = 6-7) + (5-3) = 3

For A_3 & A_8 = (5-7) + (5-3)

Minimum distance:

We have to find the minimum distance between all the points and make our first cluster.

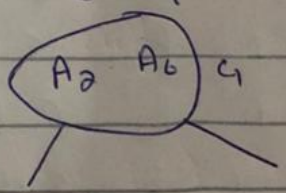
Distance Metrix

	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}
A_1	0	2	8	9	10	3	9	12	12	14
A_2	2	0	6	7	8	1	7	10	10	12
A_3	8	6	0	7	4	5	1	4	4	6
A_4	9	7	7	0	3	8	8	7	9	7
A_5	10	8	3	3	0	7	5	4	6	4
A_6	3	1	8	4	7	0	6	9	9	11
A_7	9	7	3	2	5	6	0	3	2	5
A_8	12	10	7	5	4	9	3	0	8	2
A_9	12	10	9	1	6	9	3	2	0	2
A_{10}	14	12	7	6	4	11	5	2	2	0

The smallest distance is '1' which is between "A_2 & A_6"

Merging clusters:

Merge the closet point to a cluster



$C_1 = (A_2, A_6)$

The centroid of C_1 is calculated by

$$x_{centroid} = \frac{x_1 + x_2}{2}$$

$$y_{centroid} = \frac{y_1 + y_2}{2}$$

$$x_{centroid} = \frac{2+3}{2} = 2.5$$

$$y_{centroid} = \frac{8+8}{2} = 8$$

$$= C_1 (2.5, 8)$$

Updated Matrix:

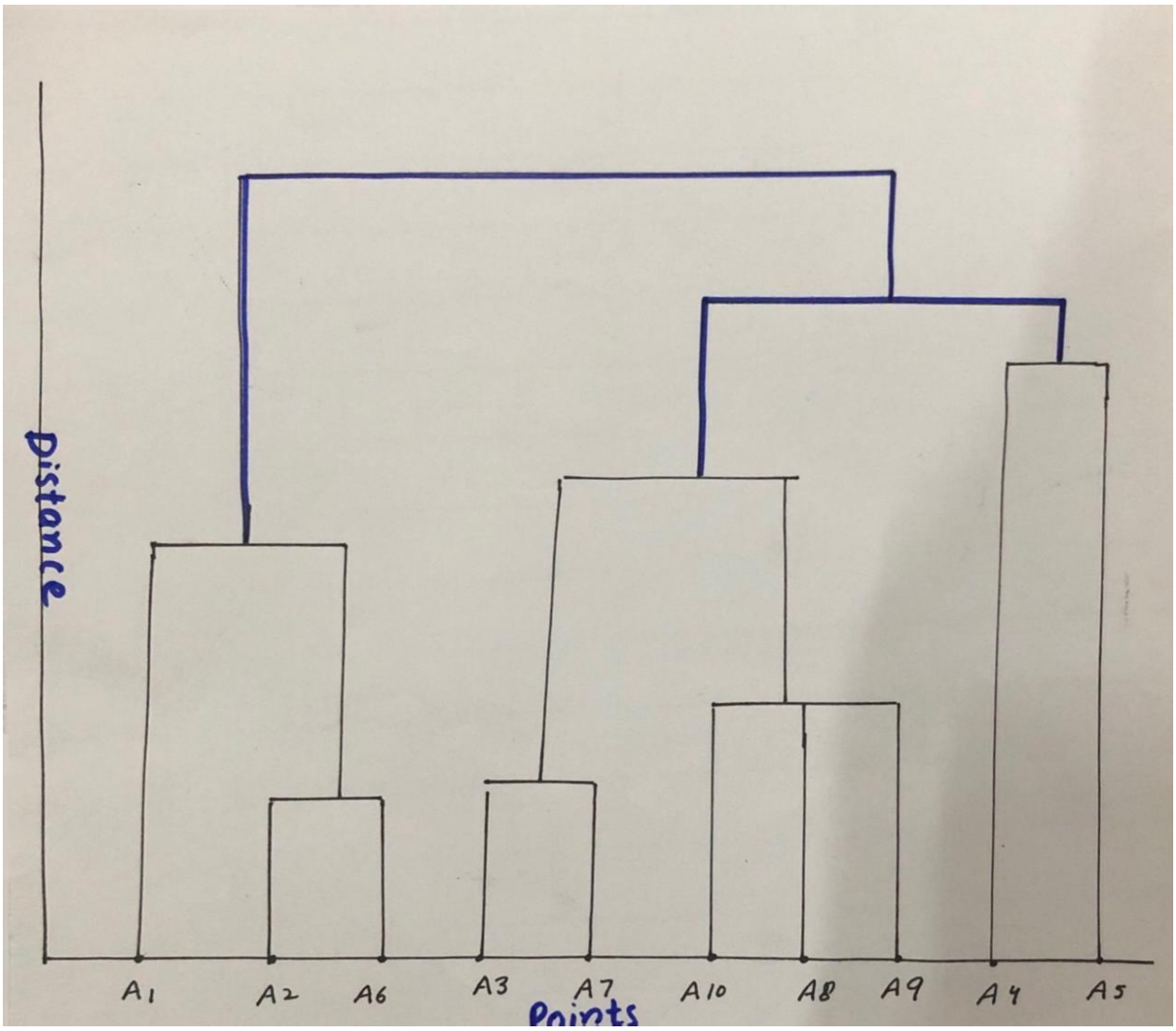
	A_1	A_3	A_4	A_5	A_7	A_8	A_9	A_{10}
C_1	2.5	5.5	7.5	7.5	6.5	9.5	9.5	11.5

Minimum distance:

Now the smallest distance is 2.5 between C_1 and A_1. So, merging A_1 into C_1

Further Process:

Now, we update the cluster and repeat the whole process and the final dendogram is made.


Distance
A1
A2
A6
A3
A7
A10
A8
A9
A4
A5
Points

Q.No.2:

Q.No.2:

Imagine we have following datasets

P_1 (2,10)
P_2 (2,8)
P_3 (5,5)
P_4 (1,2)
P_5 (4,2)
P_6 (7,3)

Methoddology:

It is basically opposite of previous techniqye, we make a larger cluster first.

step 1:

Strat with all points in one cluster first.

$C_1 = (P_1, P_2, P_3, P_4, P_5)$

step 2:

Finding pairwise distance and identify points or groups of farthest part.

Step 3:

Divide the clusters into two parts.

$C_1 = (P_1, P_2, P_3)$

$C_2 = (P_4, P_5, P_6)$

Step 4:

Repeat the process for each sub-cluster

$(P_1, P_2, P_3) \longrightarrow (P_1, P_2)$ and (P_3)

$(P_4, P_5, P_6) \longrightarrow (P_4, P_5)$ and (P_6)

Step 5:

(P_1), (P_2), (P_3), (P_4)

(P_5), (P_6)

Step 6:

Making Dendogram.